

Longitudinal changes in BMI in older adults are associated with meat consumption differentially, by type of meat consumed.

Hypotheses regarding the role of meat consumption in body weight modulation are contradictory. Prospective studies on an association between meat consumption and BMI change are limited. We assessed the association between meat consumption and change in BMI over time in 3902 men and women aged 55-69 y from the Netherlands Cohort Study. Dietary intake was estimated at baseline using a FFQ. BMI was ascertained through baseline self-reported height (1986) and weight (1986, 1992, and 2000). Analyses were based on sex-specific categories of daily total fresh meat, red meat, beef, pork, minced meat, chicken, processed meat, and fish consumption at baseline. Linear mixed effect modeling adjusted for confounders was used to assess longitudinal associations. Significant cross-sectional differences in BMI between quintiles of total meat intake were observed (P -trend < 0.01 ; both sexes). No association between total fresh meat consumption and prospective BMI change was observed in men (BMI change highest vs. lowest quintile after 14 y: -0.06 kg/m^2 ; $P = 0.75$) and women (BMI change: 0.26 kg/m^2 ; $P = 0.20$). Men with the highest intake of beef experienced a significantly lower increase in BMI after 6 and 14 y than those with the lowest intake (BMI change after 14 y 0.60 kg/m^2). After 14 y, a significantly higher increase in BMI was associated with higher intakes of pork in women (BMI change highest vs. lowest quintile: 0.47 kg/m^2) and chicken in both sexes (BMI change highest vs. lowest category in both men and women: 0.36 kg/m^2). The results remained similar when stratifying on median baseline BMI, and age-stratified analyses yielded mixed results. Differential BMI change effects were observed for several subtypes of meat. However, total meat consumption, or factors directly related to total meat intake, was not strongly associated with weight change during the 14-y prospective follow-up in this elderly population.